

EE251 Lab 4

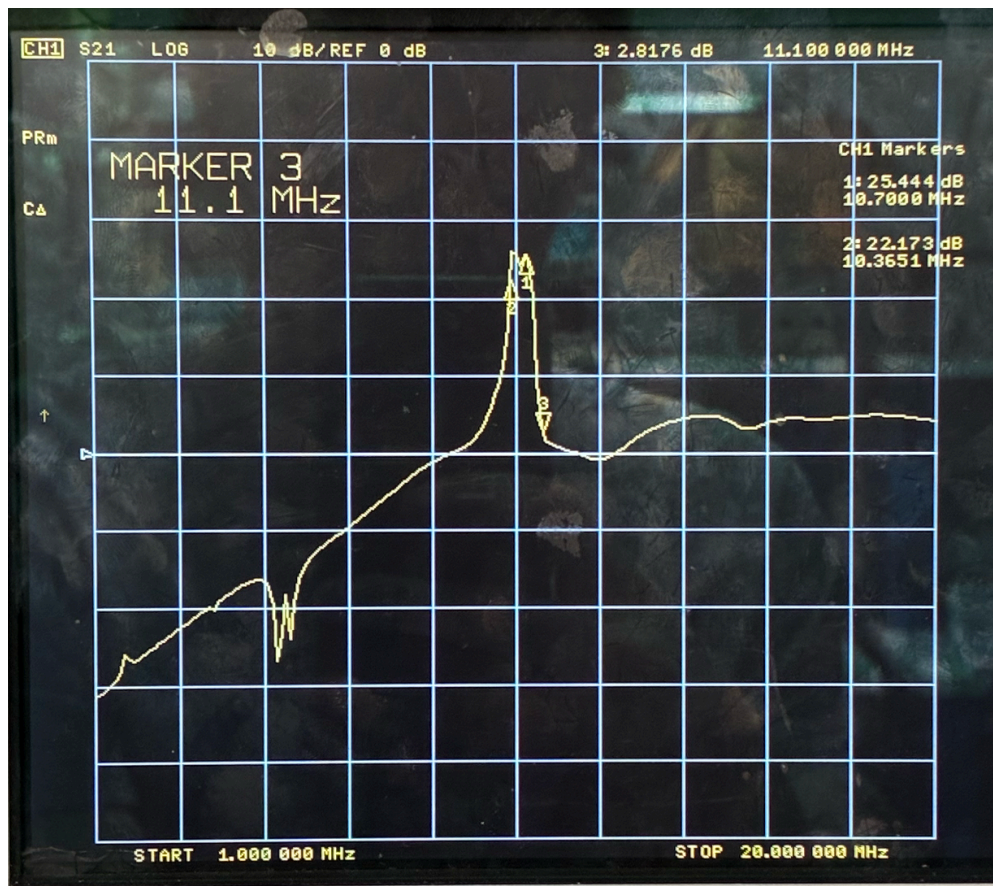
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IF Amplifier Performance

3 dB Bandwidth	Adjacent channel rejection
420 KHz	23.25 dBm

The following image is the S21 of our IF amplifier which roughly has 20dB of rejection for 400kHz off of the center frequency of 10.7MHz.



We were unable to sufficiently match our ports for the IF amplifier. The following table summarizes the S11 and S22 parameters at 10.7MHz.

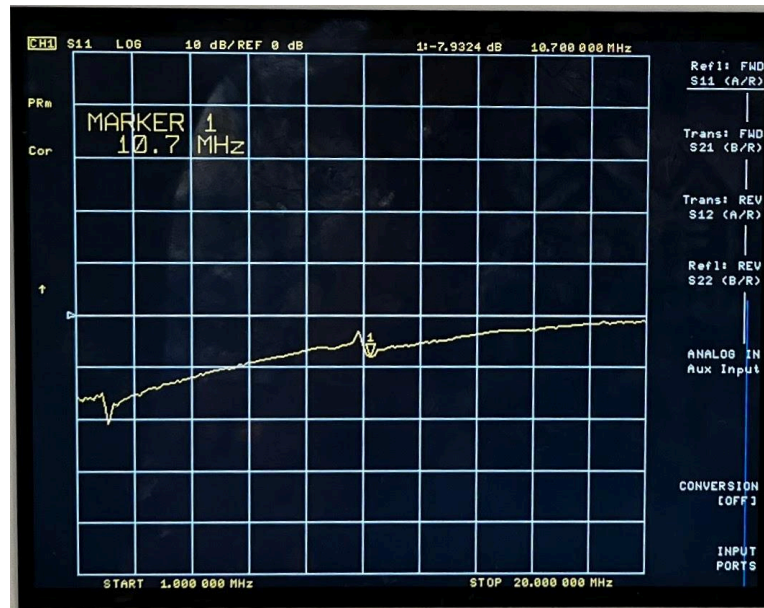
S11	-7.93 dBm
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S22

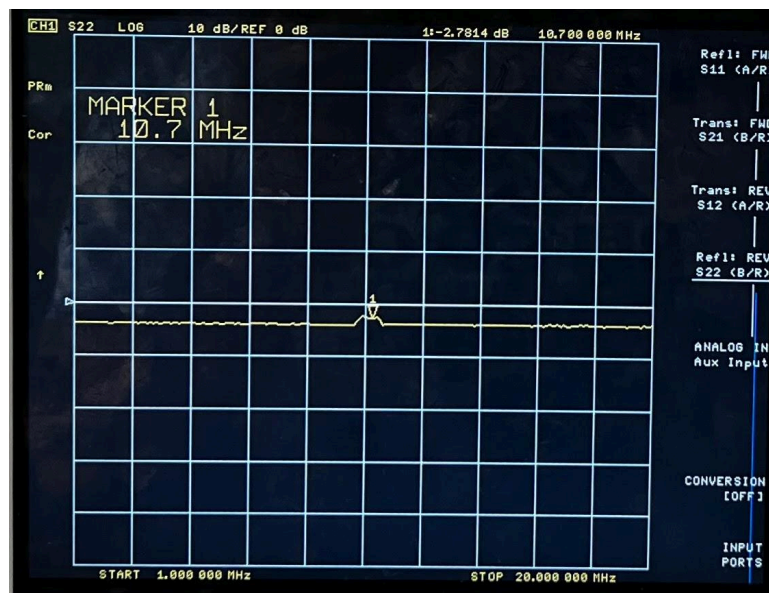
-2.78 dBm

The following images are the VNA screen captures of the ports of our IF amplifier. Going forward we need to evaluate our schematic and find ways to better match our ports.

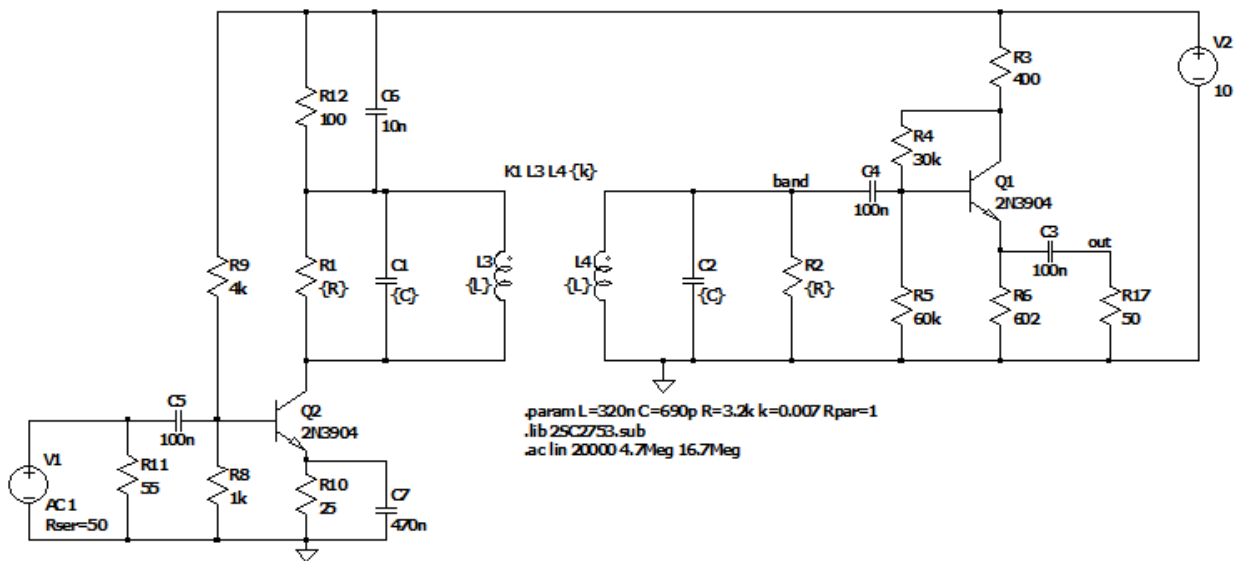
Plot of S11:



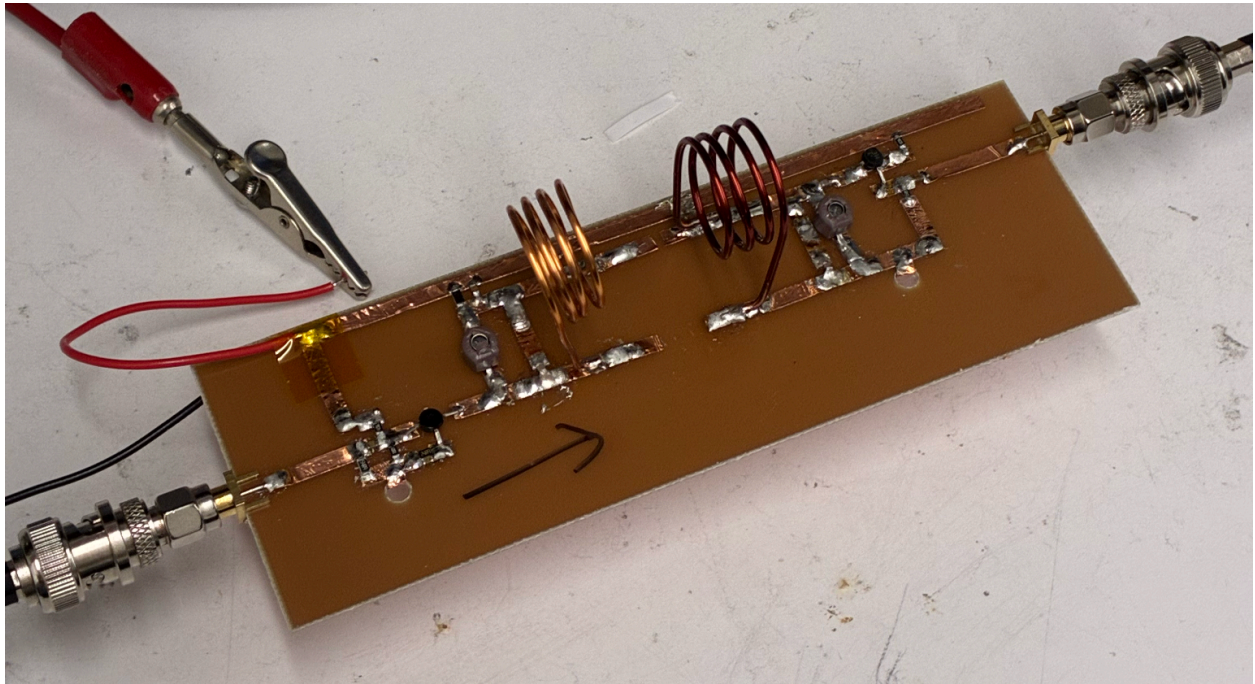
Plot of S22:



IF Amplifier Schematic



Board Picture



Technical Discussion

For our IF filter, we choose the double tuned resonator architecture to achieve a high Q for maximum adjacent channel rejection but enough bandwidth. The benefit of using a double tuned resonator compared to a single RLC bandpass filter is that we can use the coupling coefficient k as a second degree of freedom to independently adjust the bandwidth and rolloff of the filter.

Around our double tuned resonator, we added a common emitter amplifier to provide gain and act as a buffer to the input of resonator, and a common collector amplifier on the output of resonator.

Each resonator uses a $\sim 400\text{nH}$ inductor and $\sim 550\text{pF}$ capacitor. From this the resonant frequency is the following.

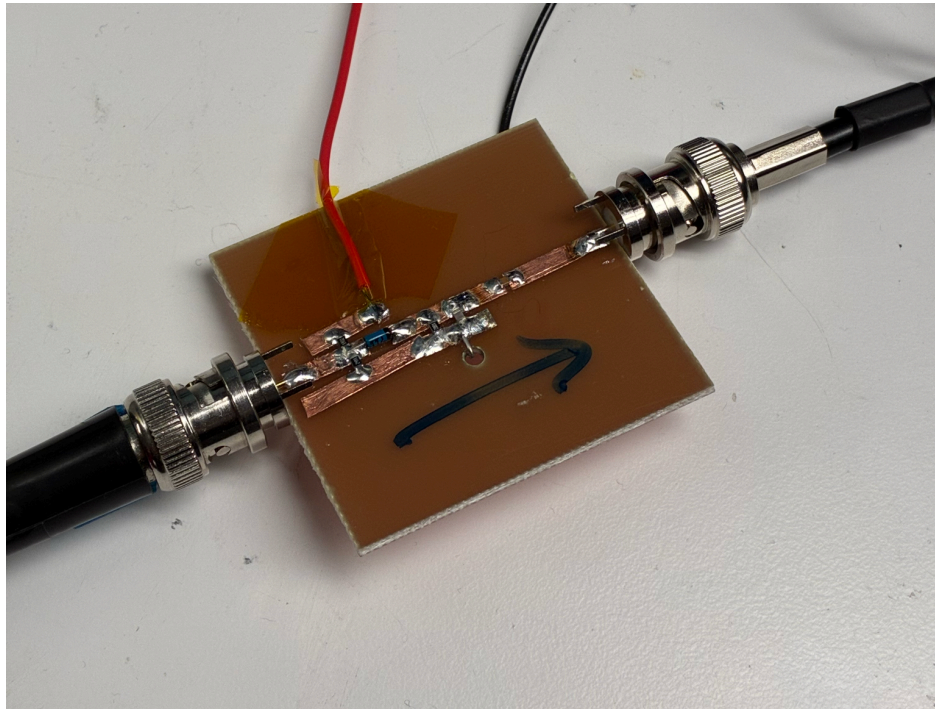
$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{400\text{nH} \cdot 550\text{pF}}} \approx 10.7\text{MHz}$$

In order to adjust our resonators so that the peaks lined up and we got sharp rolloff with a enough flatness of bandwidth, we experimented with changing the coupling coefficient k by modifying the distance between our inductors. We found that about the length of the

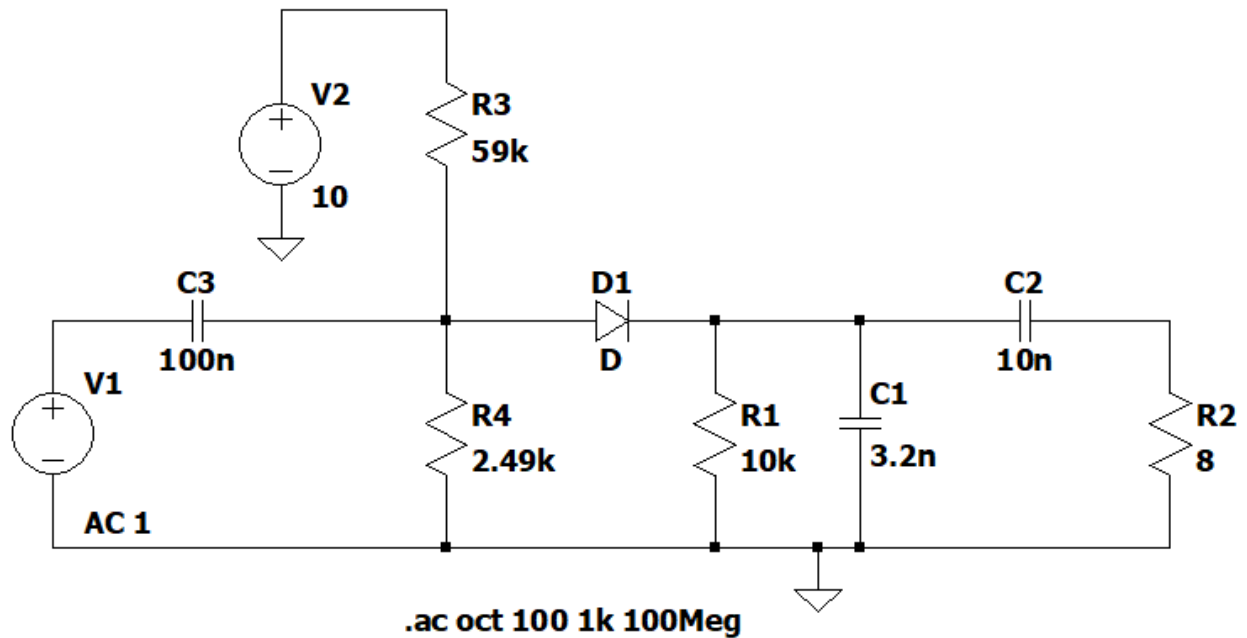
same inductor between the two inductors we built was optimal. We also added tuning capacitors in both amplifiers to fine tune the finalized bandpass filter.

In addition, in order to maintain the high Q we designed for, we needed to modify the biasing resistor values on our output amplifier/buffer in order to have a high load impedance on our double tuned filter. In our schematic R4 and R5 look in parallel at AC with a combined resistance of $(R3 + R4) \parallel R5$. At first we used the values 3k for R4 and 6k for R5, however that gave a fairly small load impedance of around 2k ohms. This caused our Q to drop and our bandwidth to expand too much and for our gain to drop. However, by increasing the values of R4 and R5 to 30k and 62k respectively, we were able to minimize the loading and preserve our high Q for optimal filter performance.

DEMODULATOR



For our demodulator, we used a schottky diode with a low turn on voltage and an output RC low pass filter to filter out frequencies above the 15 KHz limit that the human ear can hear. In addition, we biased our IF input signal around the turn on voltage of our schottky diode of 400mV so that we could receive a clean AM envelope at the output. The following is the schematic of our demodulator.



The voltage divider biases the diode barely into the turnon region and the output filter has a 3dB of 15kHz. The following is the output of our demodulator riding on top of the input AM signal.

